

Fixing the bootloader

Now we have to fix the bootloader. In this article, we look at only GRUB, since it is the most popular and compatible choice. GRUB is installed to a disk using the command `sudo grub-install DEVICE-NAME` (not the partition name) and updated using `sudo update-grub`.

One common pitfall when fixing the bootloader from a live CD is to run `sudo update-grub` directly. There are no notable errors, but the bootloader does not get updated. Has the command worked? Yes, in only that it has updated the GRUB installation of the live system.

If `update-grub` will fix the bootloader in the disk related to the running OS only, how do we update the GRUB in the hard disk or SSD without being able to boot it first? We have some solutions.

- Run the GRUB utilities provided by the live system, but instruct them to consider a different target root.
- Use `update-grub` provided by your permanent OS itself (i.e., on HD/SSD), but run it from the live CD using `chroot`.
- If your permanent disk contains any other GNU/Linux installation that is bootable, load it and run `update-grub` from it.

I don't prefer the first option since there can be version incompatibilities. The third option is the easier one. But that's not always available, so we have to go for the second.

Assuming you are running a live, rescue or different OS installation, and calling the newly copied OS (forget the old disk or partition) *target OS*, this is what you have to do.

1. Mount the root of the *target OS* to your live system (replace X and Y with the disk and partition numbers of the target OS root):

```
sudo mount /dev/sdXY /mnt
for i in /dev /dev/pts /proc /sys; do
```

```
    sudo mount --bind $i /mnt$i
done
sudo chroot /mnt
```

Now that you are in the *target OS*, issue the following commands:

```
grub-install /dev/sdX
update-grub
exit
```

It's time to restart. If you are going to issue the `umount` command (which is unnecessary), be sure to do that in reverse order (i.e., `/sys` first).

⚠ Caution: The package `grub-pc` is used for legacy booting and `grub-efi` is used for UEFI booting (names may vary slightly). Make sure the version you are using is compatible with your motherboard setup. Consult your package manager to know which version is in charge.

💡 Tip: There are several pitfalls related to UEFI. Here is a good read to start with: https://help.ubuntu.com/community/UEFI#General_principles

Cloning a disk or a partition

Taking a bit-by-bit copy may sound like something that happens only in an advanced lab. It does, but that doesn't mean you can't do it. In fact, all it takes is a single line of command:

```
cp SOURCE DESTINATION

...or:

dd if=SOURCE of=DESTINATION
```

Both *SOURCE* and *DESTINATION* can be real disks, partitions or image files. Some applications include taking the backup of a DVD or *applying* the ISO image of a live GNU/Linux distro to a CD or USB Flash drive.

This example copies Partition 1 of the first [hard] disk to Partition 2 of the second:

```
dd if=/dev/sda1 of=/dev/sdb2
```

`dd` provides more options like defining the buffer size, I/O synchronisation and error control. See *man dd* to know more.

Regardless of the tools you choose, it's a good idea to issue the `sync` command after the copying to flush any cache.

⚠ Caution: Neither *SOURCE* or *DESTINATION* can be a mounted disk or partition.

Disclaimer: All the facts and instructions given in this article are shared in the hope that they will be informative and useful. How-to parts have been tested and verified by the author. However, the author or the publisher cannot guarantee that they will work in all situations or on all machines. Things can fail due to multiple reasons including version differences, software bugs, hardware issues and human errors. Since failures during system cloning can render the system useless or even wipe-off disks, users are advised to double-check the information and the commands they issue, and then proceed at their own risk. **END** 🐧

 By: Nandakumar Edamana

The author is a developer and technology writer who primarily focuses on software freedom, privacy and other ethical issues. He has authored a couple of books and several articles for mainstream publishers. A GNU/Linux user for more than ten years, he releases his software products under free software licences, which are available on nandakumar.co.in.